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وزارة التعليم

الجامعة الإسلامية

كلية الحاسب ونظم المعلومات

قسم تقنية المعلومات

IU HOUSING SYSTEM



Student:

Ahmed Fahad Al- Harbi	421014553
Mamdouh Salah Al- Badrani	421009883
Naif Ahmed Al- Sharif	421014363

Supervisor:

Dr. Oussama Benrhouma

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Signed: _____

Stud ID.: 421014553 Stud Name: Ahmed Fahad Al-Harbi



Stud ID.: 421014363 Stud Name: Naif Ahmed Al-Sharif



Stud ID.: 421009883 Stud Name: Mamdouh Salah Al-Badrani



Date: 01/12/2024

Certificate

It is certified that project report has been prepared and written under my direct supervision and guidance. The project report is approved for submission for its evaluation.

Dr. Oussama Benrhouma

ABSTRACT

This project aims to transform the Islamic University's dormitory system into an electronic platform. It demonstrates how the system was created, reviews the system's development mechanism, and its key functions, such as login and task management. It also aims to improve the efficiency and effectiveness of dorm management, the room reservation system, record keeping, and billing. It offers significant benefits to both students and administrators, such as reducing paperwork and streamlining operations.

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Chapter One

1. Introduction

(University Housing System) is a service provided by the university to support students, aiming to facilitate the process of obtaining university housing at the best prices with a variety of services.

Currently, the administration responsible for university housing relies on a paper system in all its transactions and there is no digital system that manages all the tasks of reservation, renewal, etc. Therefore, with the electronic development, there must be a system that helps students accomplish their tasks and provide services in a greater and more efficient manner.

This service allows students to book university housing at and provides them with many services such as:

Request housing cleaning: the ability to request regular or as needed cleaning services.

Evacuation: facilitating the process of evacuating housing at the end of the residence period.

One of the most important features of the service is that the housing is located on campus and close to colleges, which provides students with comfort and easy access to study places. We will provide a powerful system to suit students who may not have experience using web pages, in order to achieve our goals and ensure efficiency and speed.



Figure 1 Digital transformation

1.1 Problem Statement

The main problem with the current system is that it relies on paper: The report focuses on solving an existing problem or what we call a loophole in (the Islamic University in particular), which is the lack of a housing system for students and the reliance entirely on paper documents and contracts.

which leads to other **sub-problems, including:**

Lack of priority, as the first to come gets housing, and therefore there is no priority for students with limited income, and students who live in remote areas. In addition, contracts can be damaged or lost.

The main challenge in moving from a manual system to an electronic system will be in improving the process of requesting rooms, and submitting applications, among other processes. The report also focuses on the most important aspect, which is related to the customer experience by providing more transparent and efficient services. It points out the problems of the current paper system: the lack of priority for low-income students and the difficulty of requesting and renewing housing. It highlights the problems that await the administration: ignorance of the university's method after graduation, as we will mention in the following examples.

- **Example of student problems:**

One of the examples that students faced was the difficulty of booking and that everything was done manually and required a lot of papers to be filled out and the difficulty of renewal in each semester and there was no priority in terms of distance and there was a condition that the student be from outside the city.

- **Example of management problems:**

- Among the problems that the administration faces: After the student graduates, the administration does not know that a party has been cleared through him, and this is one of the big problems.



Figure 2 Current problem

1.2 Goal

Our project aims to:

- Convert the traditional (manual) system to an electronic system.
- Give priority to students with limited income.
- Develop an easy-to-use and intuitive website for housing management that can be accessed by students and administrators.
- Provide accurate information on housing occupancy, room allocation, billing.
- Provide a scalable system that can adapt to future renovations and growth.
- Improve the overall student experience by providing a simplified and efficient housing management process.

Here we will mention how to achieve these goals:

1. Change the manual system into an electronic one

- How to implement: We will use modern web development technologies like PHP and JavaScript along with MySQL database to develop the e-housing management system.

2. Students with low incomes should be given priority

- How to implement: We will use to develop an algorithm of booking priorities according to financial position, taking the income data from students on specific forms.

3. Make the website easy to navigate and intuitive

- How to implement: We will to conduct usability tests with a group of students and administrators to ensure that the design is understandable and shall be easily accessible for different levels of computer skills.

4. Display correct information about housing occupancy

- How to implement: We will be to establish a system for periodic updates, integrate it with a central database where access to data is easily made.

5. Scalable system

- How to deliver: We will to design the system in such a way that it will be flexible; extra features could be added, or capacity increased without deteriorating performance.

6. Enhanced total student experience

- How to deliver: We will to be constantly soliciting for students' feedback through periodic surveys to determine what we should be working on to make the system even better in further developments.

1.3 Objective & Motivation

The project objective is to create a university housing service that meets the needs of students from outside the city.

The main objectives focus on:

- **Enhancing the ease of use of the system:**
Creating a flexible and effective system that suits the different levels of experience of students, to ensure efficiency, accuracy and speed of managing housing services.
Reducing the occurrence of problems related to housing reservation, evacuation, maintenance.
- **Continuous improvement:**
We aim to improve the quality of the application. to avoid errors and problems to achieve high efficiency in order to adapt to.
- **potential changes such as:** increasing the number of rooms or building a new floor or housing unit.
- **The motive:** is to put students studying the same major or academic level in one room or the same wings, which will help them achieve success and study and provide a housing environment that is more compatible with the needs of students.

1.4 Methodology

The methodology we will be following is Agile. Agile means flexibility, and it is based on developing projects in an incremental and iterative manner so that the team can adapt to changes faster and improve the application based on continuous feedback from users and employees.



Figure 3 Agile Development

1.4.1 How to apply the methodology in our project

The project was divided into short tasks (sprint) which means defining a specific scope of work for each (sprint) so that it can be completed within a specific period of time:

Sprint 1- Developing the Home page interface.

Sprint 2- Developing the services interface.

Sprint 3- Develop all interfaces.

Sprint 4: Testing and Feedback Implementation

Sprint 5: Final Refinements and Deployment

1.4.2 Benefits of using it

- **Adaptability to changes:** It allows you to easily make changes to the project based on feedback, ensuring that the system meets the user's needs.
- **Reducing risks:** Developing and testing small parts of the system enables problems to be identified and solved quickly.
- **Improving communication:** Weekly meetings and frequent planning help improve communication.
- **Review:** User feedback helps to know the development of the system and to know future problems.

1.4.3 How to start using Agile methodology in our project

- **Gather the team and define roles:** Our team was gathered and working together collaboratively.
- **Define the requirements list:** Gather all the functional and non-functional requirements of the system and use the user story
- **Plan the sprints:** Identify the tasks to work on in each sprint based on priority and available time.
- **Start work and use weekly meetings:** Start work on the tasks identified in the first sprint and use weekly meetings to check progress.
- **Gather feedback and repeat the process:** After each sprint, we gathered feedback from users and re-planned the next sprint based on this feedback.

1.5 Ethics of Project

In developing and implementing our University Housing System project, ethical considerations play a vital role in ensuring that the system serves students in a fair, responsible and transparent manner. The following ethical principles guide the project:

- **Confidentiality Agreements:** All employees and those involved in the development and management of the housing system will sign confidentiality agreements to prevent unauthorized disclosure of student information.

- **Transparency of Allocation:** Housing allocations will be made transparently based on clear and objective criteria, including assessment of financial need and prioritization based on proximity to the university. **Currently, the** customization process is manual, and will remain so during the initial phase of the system, with future plans to automate the customization task.
- **System Integrity:** The project will maintain high standards of integrity in the design and implementation of the system, to ensure that the system operates as intended without manipulation or errors.
- **Student Support:** We will provide adequate support for students using the housing system. Support services will be available and responsive to ensure a positive user experience.

1.6 Gantt chart

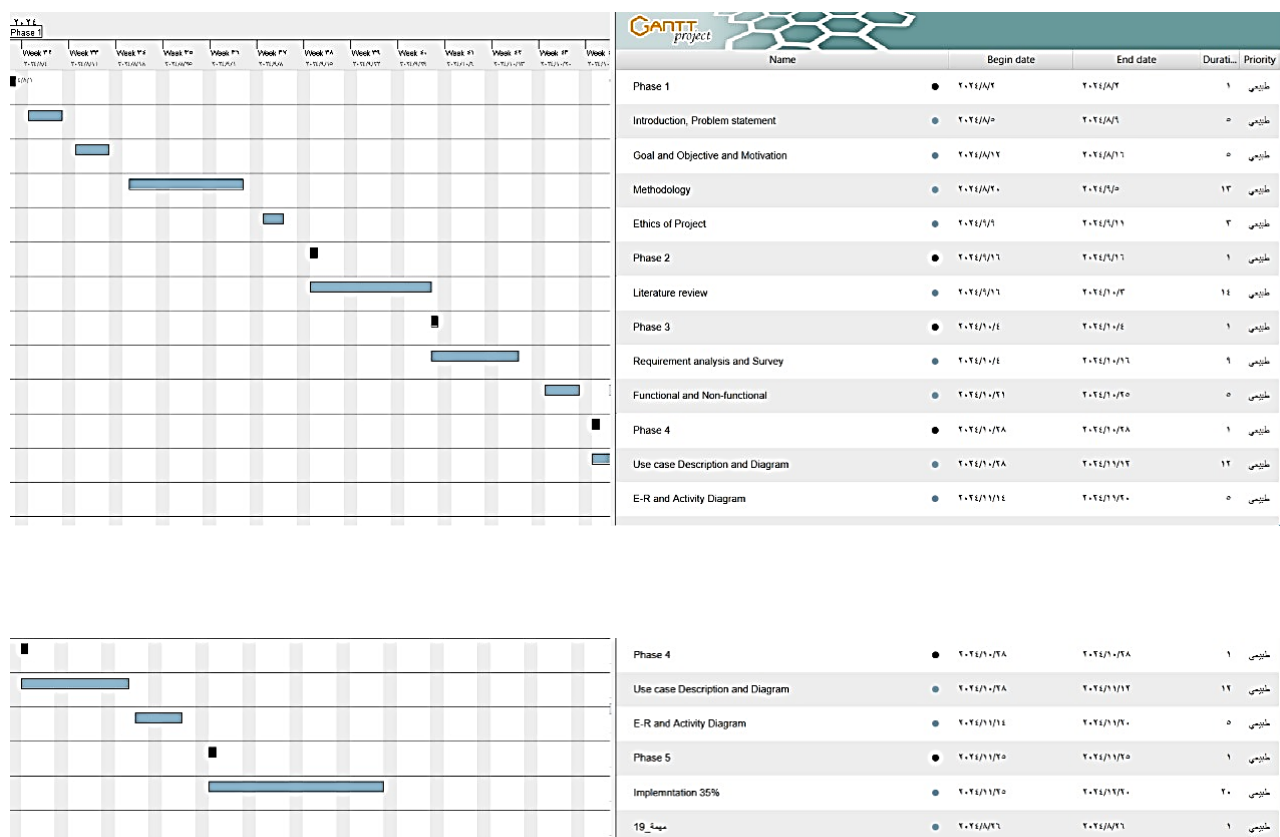


Figure 4 Gantt chart

Figure 4 refer to Gantt Chart is a visual representation to the project that serves as planning and scheduling. It is a chart that illustrates the several types of tasks or phases involved in the project and the time required for each phase.

How it works:

There is a horizontal bar representing each task/phase in the project.

The bar begins at the start of the task and ends at the end of the task.

The length of the bar corresponds to the time required to complete the task, and the color of the bar shows you if it's completed or still in progress.

In the example shown:

Phase 1: Establishing the Problem and Project Objectives.

Stage 2: Evaluation of the existing studies or evidence.

Stage 3: Examining the need and Researching.

Phase 4: Sketching diagrams that will assist in the explanation of how the systems will work.

Step 5: The realisation of the project.

In other words, the Gantt Chart keeps track of how far the project has come, by listing each task on the list and the time that it is slotted for.

Chapter Two

2. Literature View

2.1 Abstract

These research cover, but not limited to, building electronic booking and management systems of student housing facilities using state-of-the-art web technologies with the objective of enhancing educational institutions efficiency. These traditional systems are slow, error-prone and obsolete; they have been replaced with digital Systems that allows users (students & staff) to create bookings instantly, track reservation system effortlessly and this data can be accessed anytime and anywhere.

The first study: "Development of Electronic Housing Management System for Students" where a web option system was designed that accepted student meal reservation requests and provided immediate data update. With the help of the system, it was able to save time and money while allowing for faster access and a higher accuracy rate [1].

The second study: involved creating an online student housing booking application with the waterfall method in which system is implemented sequentially. This system is designed to ease the way people reserve a place and show them clearly about date range, time slots availability, and price of each facility. The application helped in minimizing the administrative mistakes as well as providing an easy and seamless interface for users [2].

The third one: Designing and Implementing An Electronic Reservation System for an Education Institution In PHP-MYSQL Technologies. We developed an interactive system that can be used to make flexibility and security bookings over rooms. This system helped in minimizing the human errors and increased the efficiency as well as transparency in reservation management [3].

Study Four: in this study, it was the University of Ghana housing management system development. The research stated that the system helped to reduce the period for housing search and streamline the allocation and registration process, which will assist in accessing vacant rooms with their prices. The outcome indicates that the system managed to minimize the cost incurred in registration as well as management processes (Okae, 2022) [4].

Study Five: It is concerning a recommendation that would create a web portal to administer student residence registration. The system consists of two primary interfaces, one side for Students and the other side for Supervisors. This system helps the parties involved in process registration by providing them with minimum time and effort. The use of electronic databases and forms also plays a role in the enhancement of housing management (Podunavac & Crnjac Milić, 2019) [5]

2.2 Keyword

Application, Database, Information System Housing, Housing Management, PHP.

2.3 Challenges

The three research's faced several challenges related to traditional systems, such as:

Human problems resulting from dealing with the paper system.

Inability to access data immediately.

Difficulty in managing large data related to reservations for university housing.

Efforts related to reservations or renewals, whether by payment or confirmation of reservations.

Thanks to the use of electronic systems via the web, these challenges were overcome, as these systems provided:

The ability to access data immediately from anywhere, anytime and with ease.

Improving transparency and accuracy of information, and reducing errors.

Managing large data easily, which helped organize reservation processes more efficiently [1] [2] [3].

2.4 Results

When comparing previous studies with our project that aims to develop a housing management system based on a web page, we will compare it between: Orientation and objectives: We found that the objectives of previous research share the objectives of our project in improving the efficiency and development of university housing management, but most previous studies focus on traditional systems and old technologies. In contrast, our project focuses on using web page technology to facilitate the reservation process and link it directly to the university website. We found a difference in the use of technology and methodology: we used the (agile) methodology and in one of the research projects (waterfall) While previous studies use traditional database management techniques, our project relies on developing an advanced web page that provides the latest data management and communication technologies [1] [2] [3].

2.5 Subtopics

- Electronic housing management: Studies show how to develop electronic reservation management systems for university housing, allowing students and staff to easily book housing rooms [3].
- Web development techniques: Research focuses on using technologies such as PHP and MySQL to develop interactive web pages and integrated back-end systems for managing reservations [1].

- Challenges in traditional systems: Studies have discussed the problems associated with paper-based systems, such as increased data processing time, difficulties in dealing with large databases, and increased likelihood of errors [1].
- Software development methodologies: Research has demonstrated methodologies such as the waterfall model for developing systems in an organized manner, ensuring maximum utilization of available resources [2].

2.6 Literature Analysis

The literature covering these topics is rich with current trends in the field of online information systems development. One of the major studies focused on the use of an online housing management system that allows employees to submit reservation requests over the internet. In this study, an apartment reservation system with interactive web pages was built using technologies such as PHP and MySQL where users can register and make reservations.

Another study looked at the development of a web application for booking university accommodation using the waterfall method, which is a traditional sequential system development method in which each stage is completed before moving on to the next stage. This method is suitable for projects with fixed and unchanging requirements.



Figure 5 Digital system

2.7 Several websites similar to the project idea include

StarRez: is an international system for managing student housing that assists universities in offering flexible solutions for application registrations, reservation management, billing, and roommate matching. The system simplifies daily operational efficiency while reducing errors.

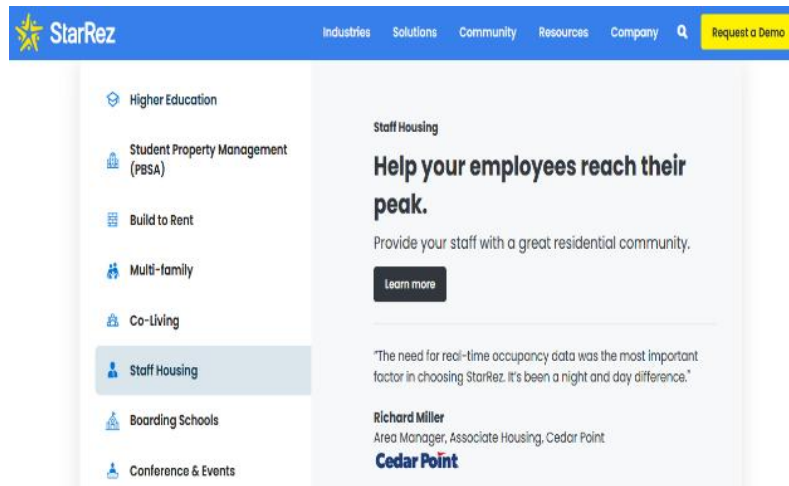


Figure 6 StarRez

Voyager: offers all-in-one solutions for student housing management, from reservation management to lease and billing setup. Using one central database can lead to higher accuracy and a reduction in manual effort.

Advantages:

Automation of day-to-day housing operations.

Accurate reporting for data analysis and management.

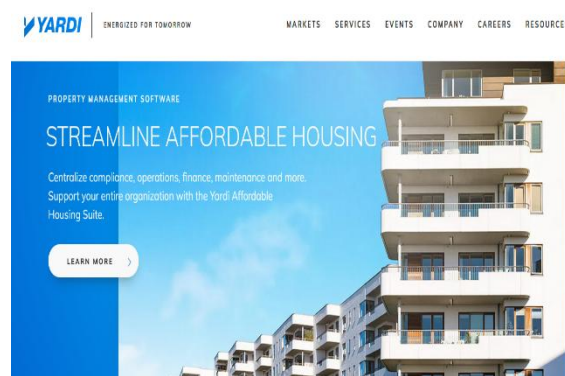


Figure 7 Voyager Student Housing by

Symlicity Residence: is a specialized system for the management and operation of student housing, designed to support universities in managing housing assignments,

tracking applications, and updating occupancy data with ease and efficiency. It provides students and administrators with an easy and integrated experience through a flexible electronic platform.

Advantages:

Room assignments according to priority or special needs.



Figure 8 Symplicity Residence

2.8 Current system

Figures 9 and 10 refer to the current housing management system at the Islamic University relies entirely on manual, paper-based procedures. All housing-related processes, such as room reservations, contract signing, maintenance requests, and student data updates, are managed through paper documents such as the poverty number and direct communication between the student and the administration. This approach creates numerous problems, such as lost or damaged documents and delays in processing applications.

In addition, there is no clear system for prioritizing housing allocations. Housing allocations are often based on a first-come, first-served basis, regardless of financial status or distance from the university. Communication between the administration and students is ineffective, and there are no diverse payment methods (only cash). There is no accurate system for tracking requests or updating maintenance. All of these challenges require a digital solution to improve the system's performance.

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Figure 9 Documentation 1

..... : الرقم

..... : التاريخ

..... : التوقيع

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



إسكان الطلاب السعوديين

المملكة العربية السعودية
وزارة التعليم
إدارة الإسكان للطلاب السعوديين

استمارة تسليم سكن

حفظه الله

الأستاذ / مشرف الوحدة الثالثة عشر (الخاصة بالسعوديين)

الإسلام عليكم ورحمة الله وبركاته، وبعد

فإن الطالب :

اسم الطالب	رقمه الجامعي		
نوع السكن	رقمه مزدوجة		
الوحدة ١٣	الدور	رقفة	السرير

يرحب به تسليم السكن الموضح أعلاه، أرجو الالتزام بإنهاء الإجراءات اللازمة واستلام الأثاث منه.

وفق الله الجميع لمقامه .

العقير التنفيذي لصندوق الطلاب

/ إبراهيم بن محمد عراقي

حفظه الله

أستاذ مع السلام للدمير التنفيذي لصندوق الطلاب

وقد تم استلام الأثاث من الطالب المذكور في الغرفة المذكورة، تكما هو موضح أعلاه.

مشرف الوحدة الثالثة عشر

..... /

أثاث الغرفة

دولاب	مراير	مكتب	كرسي	مفاح

مع السلام لمسؤول المشاريع وإسكان الطلاب السعوديين

لإستعمال المزم خلال تسليم الطالب المذكور شيكته التأمين.

العقير التنفيذي لصندوق الطلاب

/ إبراهيم بن محمد عراقي

الهيئة العامة ب (١٧٠٠) هاتف / ٨٧٧٠٠٠٠ / فاكس / ٨٧٧٧٠٠٠٠ البريد الإلكتروني / hr@ha.edu.sa

Figure 10 Documentation 2

2.9 Conclusion

These studies show that switching to electronic systems to manage reservations and accommodation at educational institutions is an important step to improve management efficiency and reduce errors associated with traditional paper systems. The use of web technologies such as PHP and MySQL and software development methodologies such as waterfall have proven to be effective in providing flexible and comprehensive solutions that meet the needs of students and staff in terms of accommodation reservations and university facilities.

Furthermore, challenges faced by traditional systems, such as slow procedures and human errors, are strong motivations for the adoption of digital solutions that can save time and effort and increase transparency in administrative processes. The findings suggest that electronic systems not only solve administrative problems, but also help improve user experience and increase satisfaction with the services provided.

In short, the introduction of an electronic appointment system has made a significant contribution to the development and improvement of university management and is a step towards the introduction of more advanced technological solutions in the future.

Chapter Three

3. Requirement Analysis

The goal of the system is to satisfy the needs of stakeholders. To satisfy them, it is important that the system provides what they expect. To effectively capture these needs, we rely on two main methods: interviews and surveys. Both methods help us gain in-depth insights into the users' opinions on the proposed system.

3.1.1 Questionnaire

A questionnaire is a tool that collects information from participants through a series of questions. It can contain written or electronic questions. The tool allows us to collect feedback from participants on existing services and their expectations for the new system. Questionnaires are an effective way to determine visitor demographics (such as age, gender, and language) and their satisfaction with current services.

3.1.2 Interviews

Interviews are an effective method of collecting data through direct communication between the interviewer and the interviewee. Interviews can be conducted in person or over the phone, allowing the interviewer to explore the interviewee's perspective in depth. In our project, we planned to conduct interviews with officials at the Prophet's Mosque to get a clear understanding of the services available and determine the specifications required for the project. We are currently facing challenges in getting approval for these meetings but hope to hold them soon.

3.1.3 Action Plan

The first phase of the needs assessment will be based on a review of similar applications and feedback from surveys and interviews with residents and visitors. We collected information from approximately 60 participants, which gave us a clear understanding of the services required by users and their expectations. Through this effort, we aim to identify system requirements, ensuring that the needs of all users and stakeholders are met. This leads to the development of an effective and appropriate system that meets their expectations.

3.1.4 Survey

We conducted a survey that yielded useful information from about 46 people. We asked common questions about the current system to see how students responded to the current system.

How do you know about the manual housing application system? **Figure 11** refer to this question aims to understand the methods students have discovered to access the application link. It appears that students learn about these methods in various ways through the information they receive, indicating a lack of official awareness of the system.



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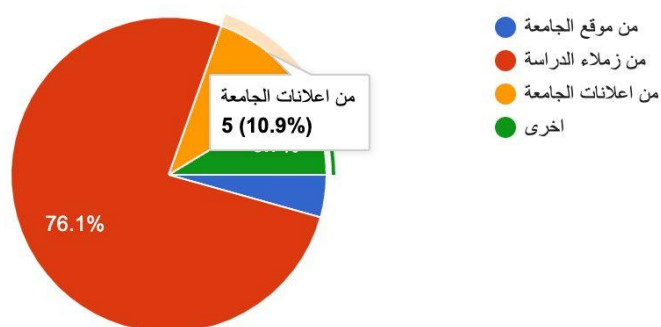


Figure 11 Survey response1

Did you encounter any problems during the application process? **Figure 12** refer to a large number answered yes, indicating challenges in accessing housing services, whether in terms of location or the required manual procedures.



هل واجهت أي مشاكل أثناء عملية التقديم؟

ردًا 46

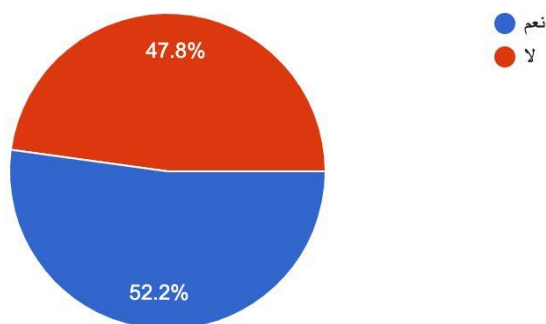


Figure 12 Survey response2

How easy is it to fill out a manual housing application? **Figure 13** refer to this question aims to gauge students' perceptions of the ease of using the current manual application system. The results showed that most students find it complex.

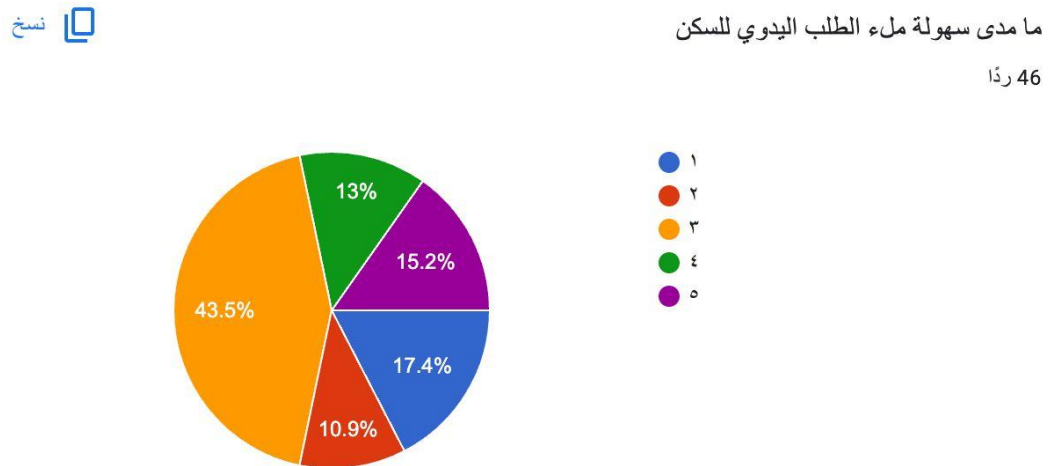


Figure 13 Survey response3

Did you receive updates on the status of your application after submitting it? **Figure 14** refer to this question aims to measure the system's interaction with students after submitting an application, in terms of responses or follow-up.



Figure 14 Survey response4

How satisfied are you with the time it takes to process your application after submitting it? **Figure 15** refers to this question measures students' satisfaction with the time it takes the administration to respond to and process applications. A large percentage of students expressed dissatisfaction with the speed of processing.



ما مدى رضاك عن الوقت المستغرق لمعالجة طلبك بعد تقديمه؟

46 ردًا

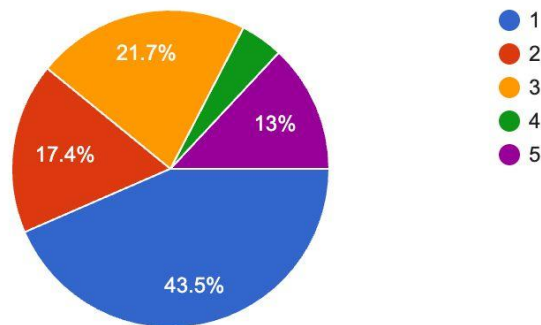


Figure 15 Survey response5

How would you rate the level of support provided by the housing administration during the application process? **Figure 16** refer to this question aims to measure the quality of support provided by the housing administration department to students during the application process. It revealed that many students indicated that it was weak or average.



كيف تقيم مستوى الدعم المقدم من إدارة السكن خلال عملية التقديم؟

46 ردًا

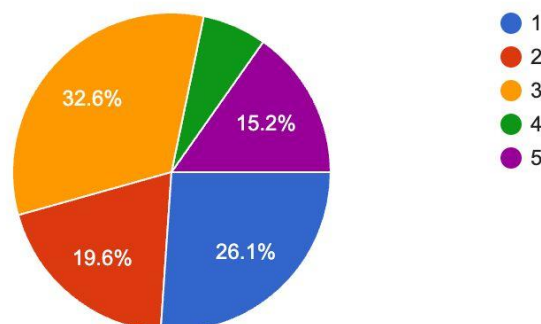


Figure 16 Survey response6

Do you prefer a manual or electronic application system? And why? **Figure 17** refer to this question aims to understand students' preferences for manual or electronic housing applications, along with reasons. The results showed that the majority prefer the electronic system, citing it as easier to use, faster, and time-saving.

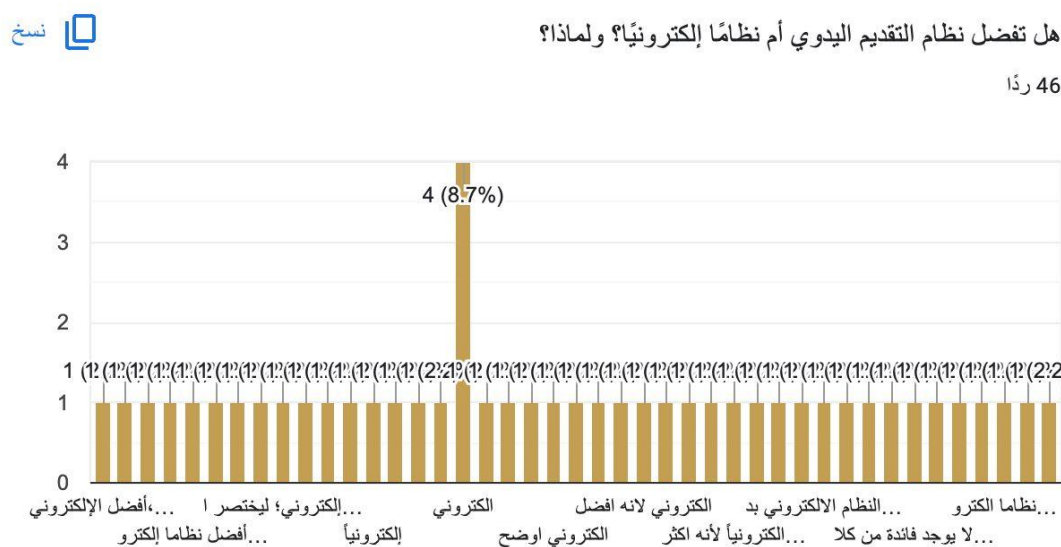


Figure 17 Survey response7

Do you have any additional comments about the manual submission experience? **Figure 18** refer to this question allows students to express their different opinions about the current manual system. Comments repeatedly indicate that the paper system causes delays and lost documents.

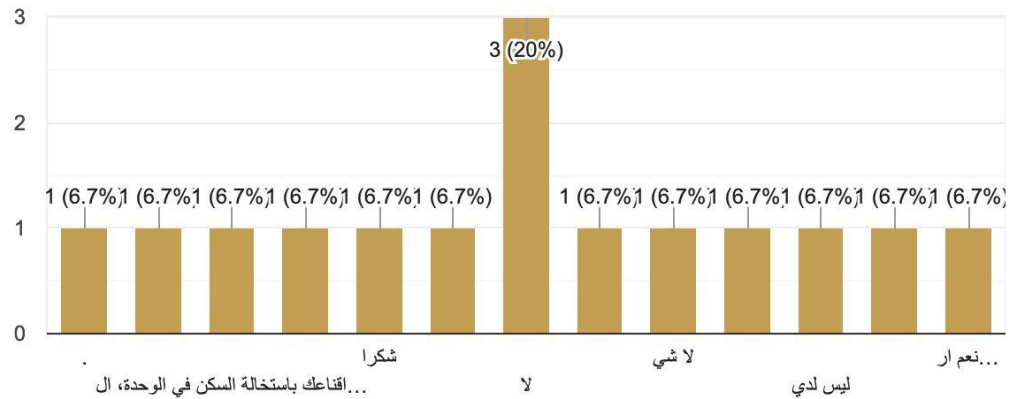


Figure 18 Survey response8

3.1.5 Analyzing the responses

The results revealed clear challenges in the manual housing management system, most notably: difficulty accessing services, delays in processing applications, lack of updates, and poor support. The responses also indicated that the majority of students prefer a digital transformation to an electronic system, which provides speed and ease in submitting and tracking applications. The results clearly indicate a need to develop a digital system that meets user expectations and improves the university housing experience, these results clearly confirm the validity of the project's approach and highlight the importance of implementing a modern digital housing system that meets students' actual needs.

3.2 Requirements

3.2.1 Introduction

This document stipulates the room reservation and maintenance management system requirements, including both functional and non-functional requirements that the system should meet to effectively realize its goals.

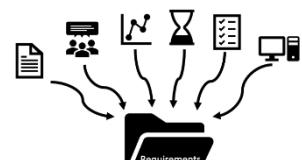


Figure 19 Requirements

3.2.2 Functional Requirements

These include account management, room booking, maintenance management, complaints processing, payment systems, report generation, among others. Each function is developed to answer the needs of users with user-friendly interfaces for accurate and reliable outputs.

3.2.3 Non-Functional Requirements

The non-functional requirements are the performance, security, and high availability. The system shall be fast, secure, and available at any time and shall support more users without degrading the performance.



Figure 20 Non-Functional Requirements

3.2.4 System Requirements

The required hardware and software for the functioning of the system are included in the document. Modern tools and technologies are identified as one of the major determinants of the:

- **Hardware requirements**

There are no special hardware requirements for running application, the smartphone or tablet is enough, and it must have network connection.

- **Software requirements**

The user must have smart device like mobile phone or tablet for using this web site.

Table 1 Functional Requirements

Functional Requirement	Description	Inputs	Outputs
Account Management	Manage user accounts (create, update, delete).	User info (name, email, password)	Account confirmation, error messages
Room Reservation	Search and book available rooms.	Search criteria (dates, room type)	Booking confirmation, availability status
Maintenance Management	Track maintenance requests.	Request details (room, issue description)	Status updates, notifications

Complaints Management	Submit and track complaints.	Complaint details (user ID, description)	Acknowledgment, status updates
Reports and Statistics	Generate reports on activities.	Data parameters (date range, user ID)	Generated reports, visual statistics
Communication	Provide direct communication channel between users and administrators		News and updates
FAQ and Support	Provide answers to common questions		User feedback and evaluation

- Implementation approach and tools used:

1. Account management:

- Implementation approach: User registration will be done along with profile management and login authentication.
- Tool to be used: Web framework - Django/Flask and Database - MySQL/PostgreSQL.

2. Room Reservation:

- Implementation approach: Design an interface through which the bookings can be made and it shows availability confirmation in a flow.
- Tool to be used: Front-end libraries (React/Angular), Backend APIs, Database.

3. Maintenance management:

- Implementation approach: Create workflows to submit, track, and resolve maintenance issues.
- Tools used: Task management tools, notification systems, and databases.

4. Complaints Management:

- Implementation approach: Design a workflow to create, track, and close maintenance tickets.
- Tools: Task management tools, notification systems, and databases.

5. Reporting and Statistics:

- Strategy: Implement complaint forms and tracking systems to check their resolution.
- Tools: Web forms, databases, and notification systems.

6. Communication:

- Implementation Approach: Implement a messaging/notification system to enable direct communication between users and administrators.
- Develop a news/update feed to display relevant information to users.
- Tools Used: Messaging/notification platform (e.g., WebSockets, Firebase Cloud Messaging).
- Content Management System (CMS) or custom-built news/update module.

7. FAQ and Support:

- Implementation Approach: Create a searchable FAQ section to provide answers to common questions.
- Implement a feedback/evaluation mechanism for users to submit comments and ratings.
- Tools Used: Knowledge base/FAQ management system.
- Online form builder or custom-built feedback system.
- Database to store FAQ entries and user feedback.

Table 2 Non-Functional Requirements

Requirement	Definition	Inputs	Outputs
Performance	The system should respond to user requests within a maximum of two seconds.	User requests	Response within two seconds
Availability	The system should be available to users 24/7.	System servers	System availability around the clock
Maintainability	The system should be well organized and commented to facilitate future maintenance and updates.	System code	Well-organized code with comments for easier maintenance
Ease of Use	The system should have an intuitive and easy-to-use interface with clear instructions.	User interface design	Intuitive interface and clear instructions

- **Performance:**

It is the capability of a subsystem or element to react quickly to the user. Against user request, the system needs to work within a duration of not more than 2 sec.

Inputs: Requests made by a user — e.g., when they click a button or enter data.

Outputs: System output within a maximum of 2 sec.

- **Availability:**

The system should always be accessible to the users at anytime and anywhere without interruption (24/7).

System Serves: The servers that control the process and allow for continuous operation.

Outputs: A 24×7 availability system.

- **Maintainability:**

The system should be simple to maintain (and update). The code must be clean and documented with relevant comments so that developers can make changes or fixes easily.

Inputs: Code for a system, constructed while the program is being developed.

Outcomes: Well-structured, annotated code that is easy to maintain and continue developing.

- **Ease of Use:**

The system should be user-friendly, with an interface that is intuitive and easy to navigate, and the instructions provided are such that users find it effortless to interact with it.

Input: Visual Design that feels minimal and not cumbersome.

Results: A simple and highly intuitive UI with no need for clarification.

Chapter Four

4. System design and implementation strategies

In this chapter we explain the design of the University Housing Management System that was used to develop it. Instead, the design phase is a major step that can convert already defined requirements to a rigid and technical layer of protection that must be appropriately implemented.

The system design minimizes resource usage, with the ability to scale if required, and it means that the system should be easy to use and flexible enough to fit additional development. Use case diagrams have been used in UML to systematically capture and present how a user is interacting with the set of components provided by the system.

Moreover, the chapter emphasizes on how we implemented these strategies such as the programming languages, frameworks and databases used in developing the system. This chapter aims to capture a broad picture of how the system behaves in order to fulfill user needs effectively, by examining use case scenarios and alternative flows.

Table 3 USE-Case description

Job Name	Room Booking
Related Requirements	User, System.
Initiating Actor	Student.
Actor's Goal	Book a room in the university housing
Participating Actors	System, Housing Staff.
Preconditions	Student has an active account.
Postconditions	Room successfully booked.
Flow of Event for Main Success Scenario	1. The student logs in and selects "Book Room". 2. The student specifies term and duration. 3. The student is assigned a room directly from the system. 4. System confirms booking and sends confirmation message.
Flow of Events for Alternate Scenario	1. If no rooms are available, the system notifies the student and suggests a different date.

Table 3 shows how a student can book a room. The student logs in, selects the service, and then the system assigns his room directly.

Table 4 Maintenance Request

Job Name	Maintenance Request
Related Requirements	User, System.
Initiating Actor	Student or Admin.
Actor's Goal	Submit or manage maintenance requests.
Participating Actors	System, Maintenance Team.
Preconditions	Student or staff is registered in the system.
Postconditions	Request is logged or status updated.
Flow of Event for Main Success Scenario	1. The student selects "Maintenance Request" from the menu. 2. The student describes the issue and specifies the room. 3. System receives the request and notifies the maintenance team. 4. The maintenance team updates the request status upon resolution.
Flow of Events for Alternate Scenario	If the request information is incomplete, the system prompts the student to complete missing details.

Table 4 shows how to request maintenance. The student requests the service and explains its details. The request is then sent to the admin, who then processes it.

Table 5 Complaint Tracking

Job Name	Complaint Tracking
Related Requirements	User, System.
Initiating Actor	Student or Admin.
Actor's Goal	Submit a complaint or track complaint status.
Participating Actors	System, Housing Administration.
Preconditions	Student has an active account.
Postconditions	Complaint is logged or status updated.
Flow of Event for Main Success Scenario	1. The student logs in and selects "Submit Complaint". 2. The student completes complaint details and submits. 3. System receives complaint and confirms receipt. 4. Housing administration follows up and updates the complaint status.
Flow of Events for Alternate Scenario	If complaint details are incomplete, the system prompts the student for missing information.

Table 5 indicates how to request a complaint. The student requests the service and explains its details. The request is then sent to the admin, who then processes it.

Table 6 Report Generation

Job Name	Report Generation
Related Requirements	User, System.
Initiating Actor	Administration.
Actor's Goal	Obtain a report on housing occupancy or maintenance status.
Participating Actors	System, Database.
Preconditions	The user has administrative privileges.
Postconditions	The report is successfully generated and provided to administration.
Flow of Events for Main Success Scenario	1-The employee selects "Generate Report" from the system. 2. The employee selects the type of report (such as works or maintenance). 3. The system extracts the basic data and displays it in the report. 4. The employee can download or print the report.
Flow of Events for Alternate Scenario	If insufficient data is available to generate the report, the system displays a message indicating that.

Table 6 indicates that the admin can create reports, including the final student report generated for him after the reservation, as well as reports on maintenance requests and complaints.

Figure 21 shows a user case diagram, showing how different entities in the system, such as the student and the supervisor, interact with the basic functions and how each entity handles the services assigned to it, based on their role within the system.

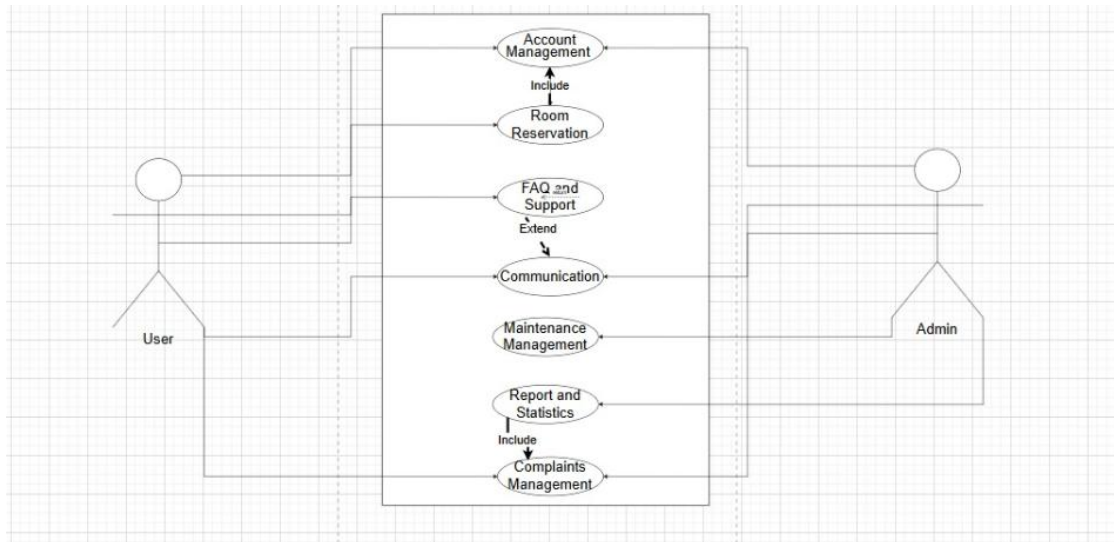


Figure 21 Use Case diagram

Figure 22 refers to a class diagram and its benefit is that it shows the main entities important for programming the system and explains the mechanism and operation of the site starting from the student and his entry into the system until generating reports and also the admin.

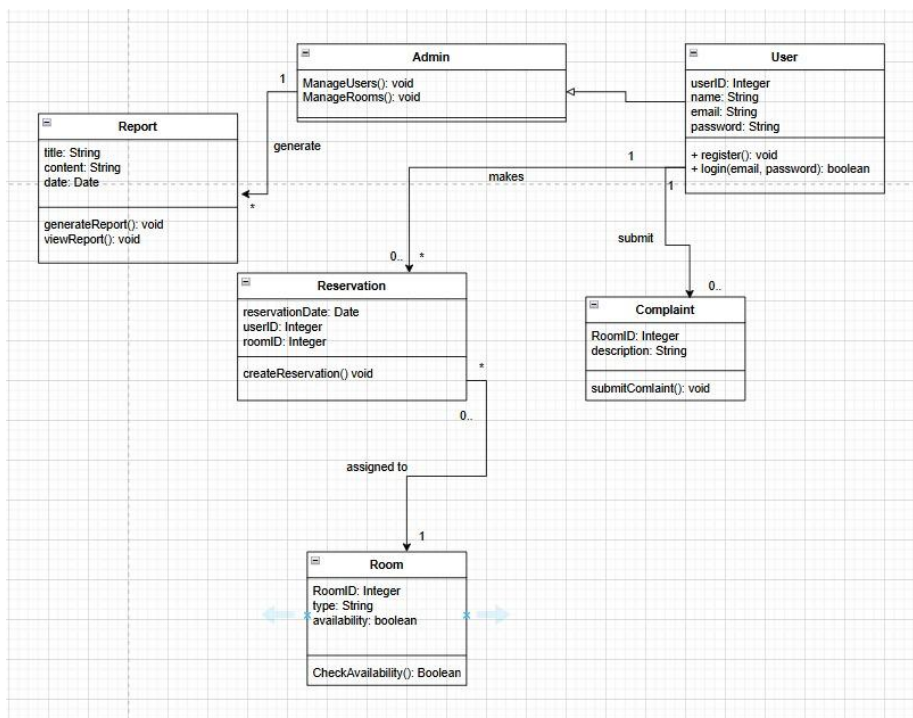


Figure 22 Class diagram

Figure 23 refers to the activity diagram, which shows how the system works from the beginning until the final step or exiting the system. Here, it is shown that the student first logs in, and then the system verifies the information. After that, the student chooses the service he wants and completes the rest of the procedures.

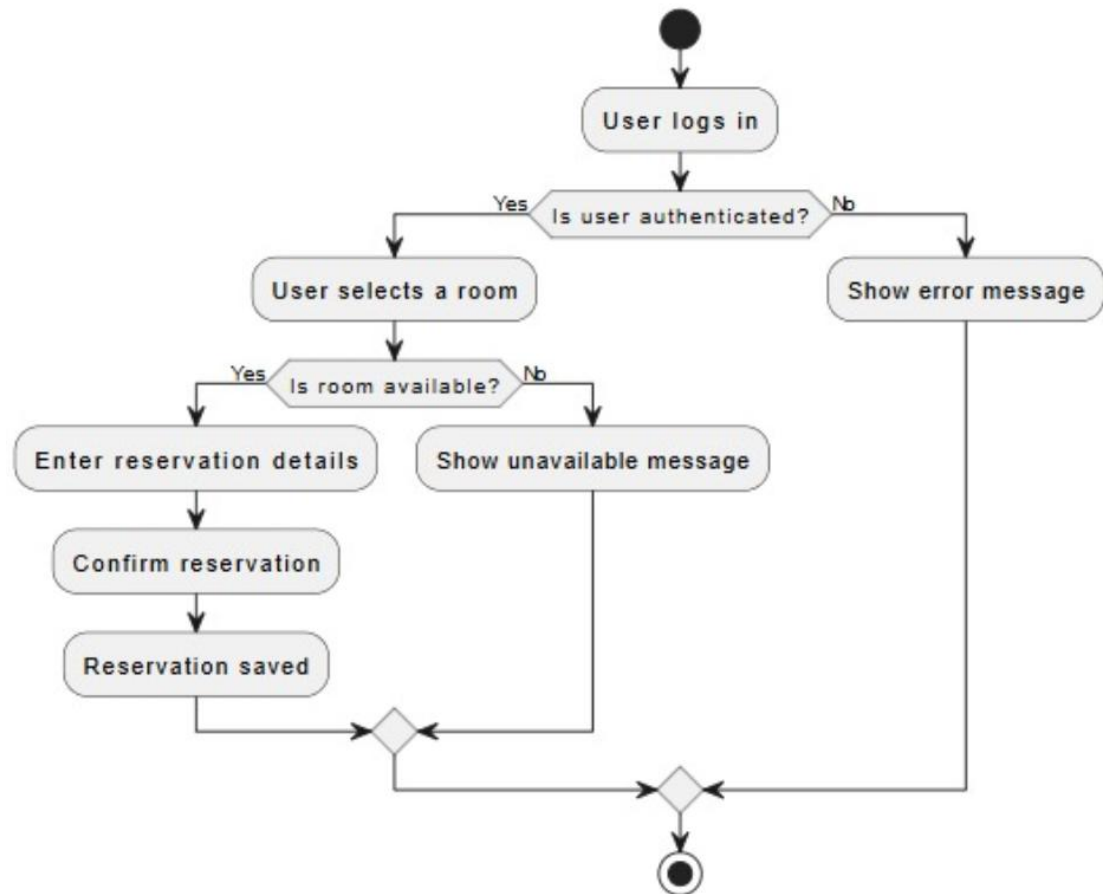


Figure 23 Activity diagram

Figure 24 refers to the maintenance request step in the sequence diagram, which is the moment the student sends a request and the system is alerted and then informs the maintenance team that the student has sent a maintenance request.

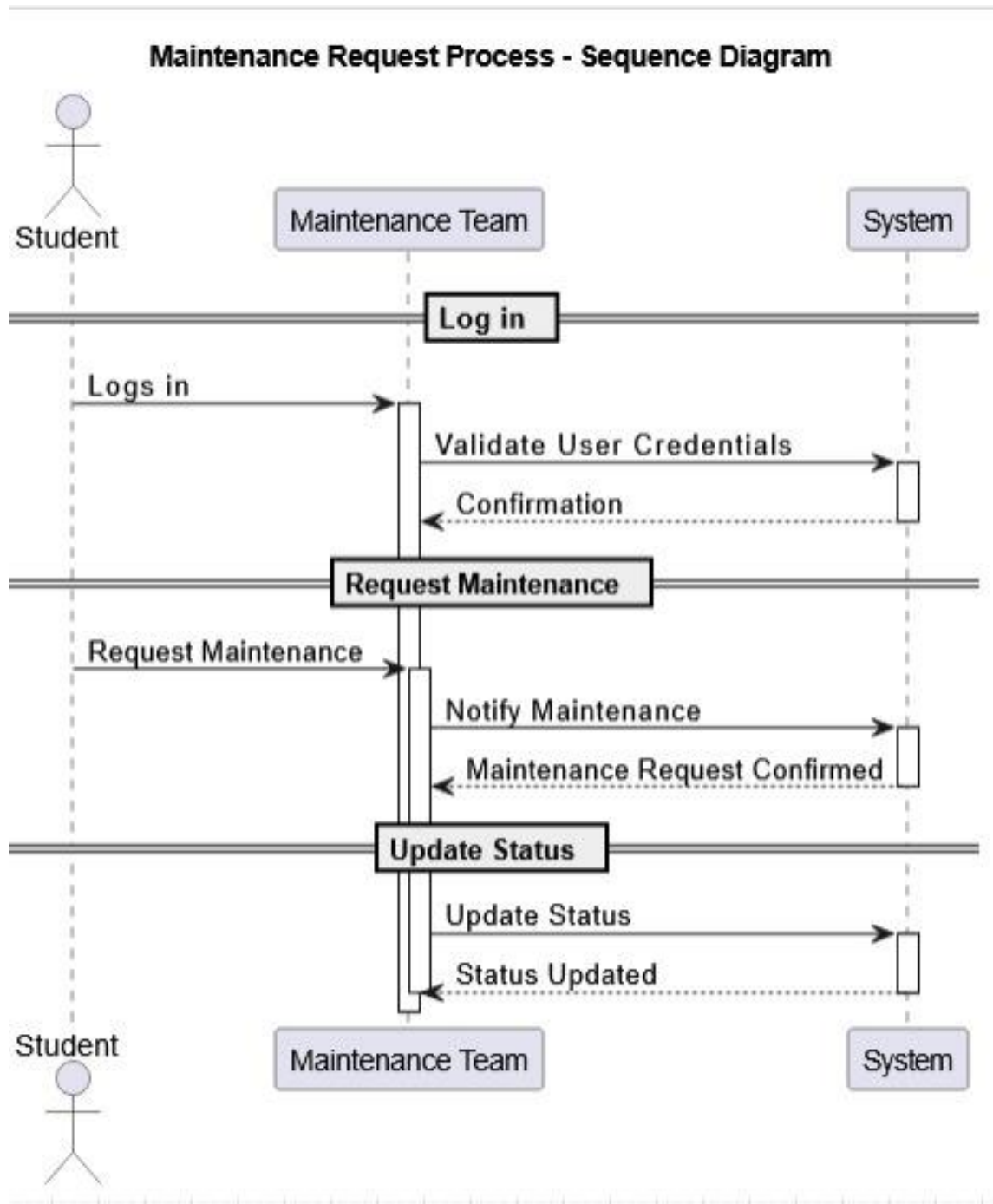


Figure 24 Maintenance Request Process - Sequence Diagram

Figure 25 refers to a sequence diagram that illustrates the complaints tracking mechanism. It includes several steps: First, the student logs in and verifies their data. They then submit a complaint, which is registered in the system. After a certain period, the administrator updates the request, which results in changes to the system.

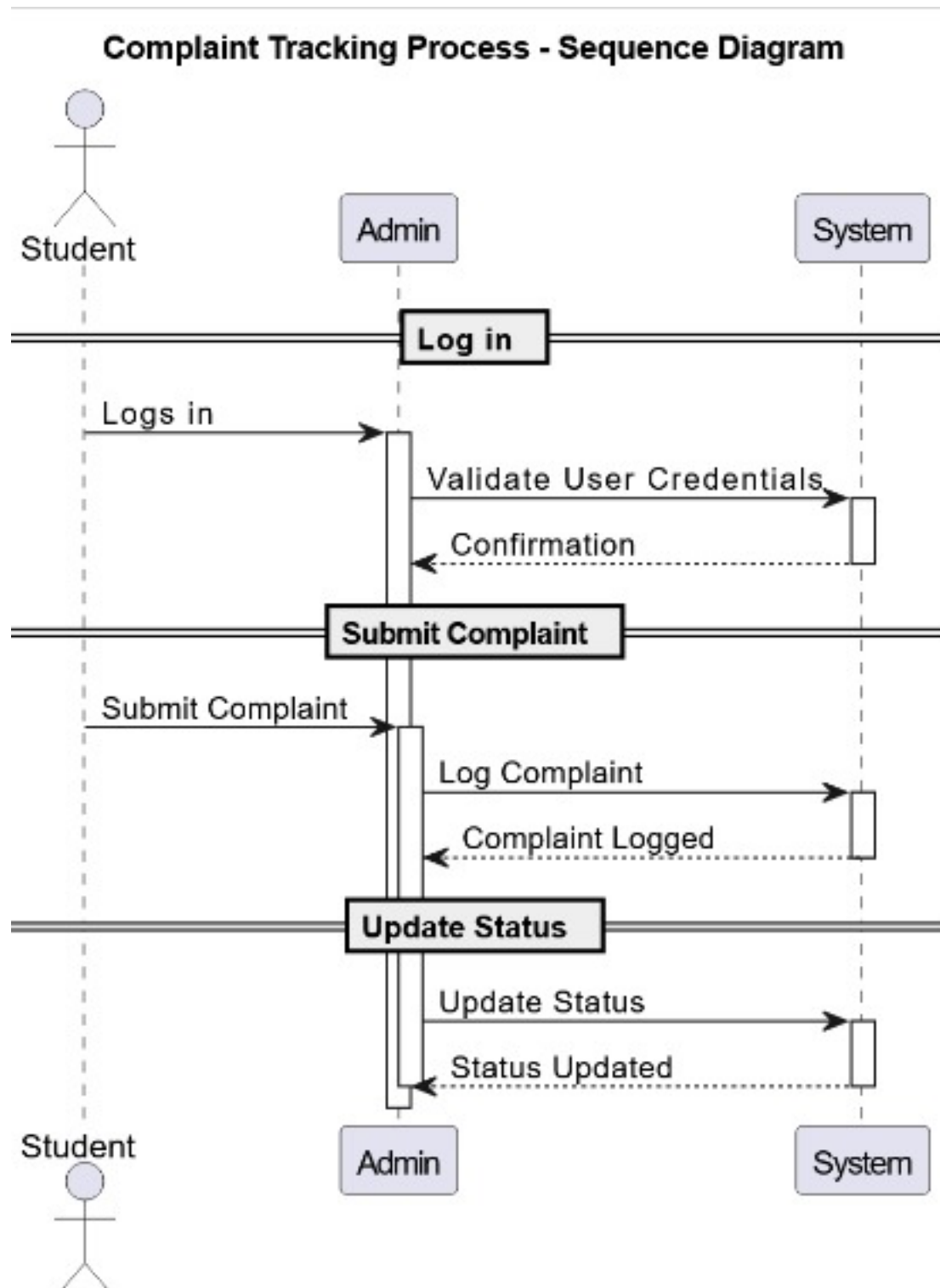


Figure 25 Complaint Tracking Process - Sequence Diagram

Figure 26 refers to how to choose a room in the sequence diagram, which is the service in which the queue books a room and the system allocates it directly to him.

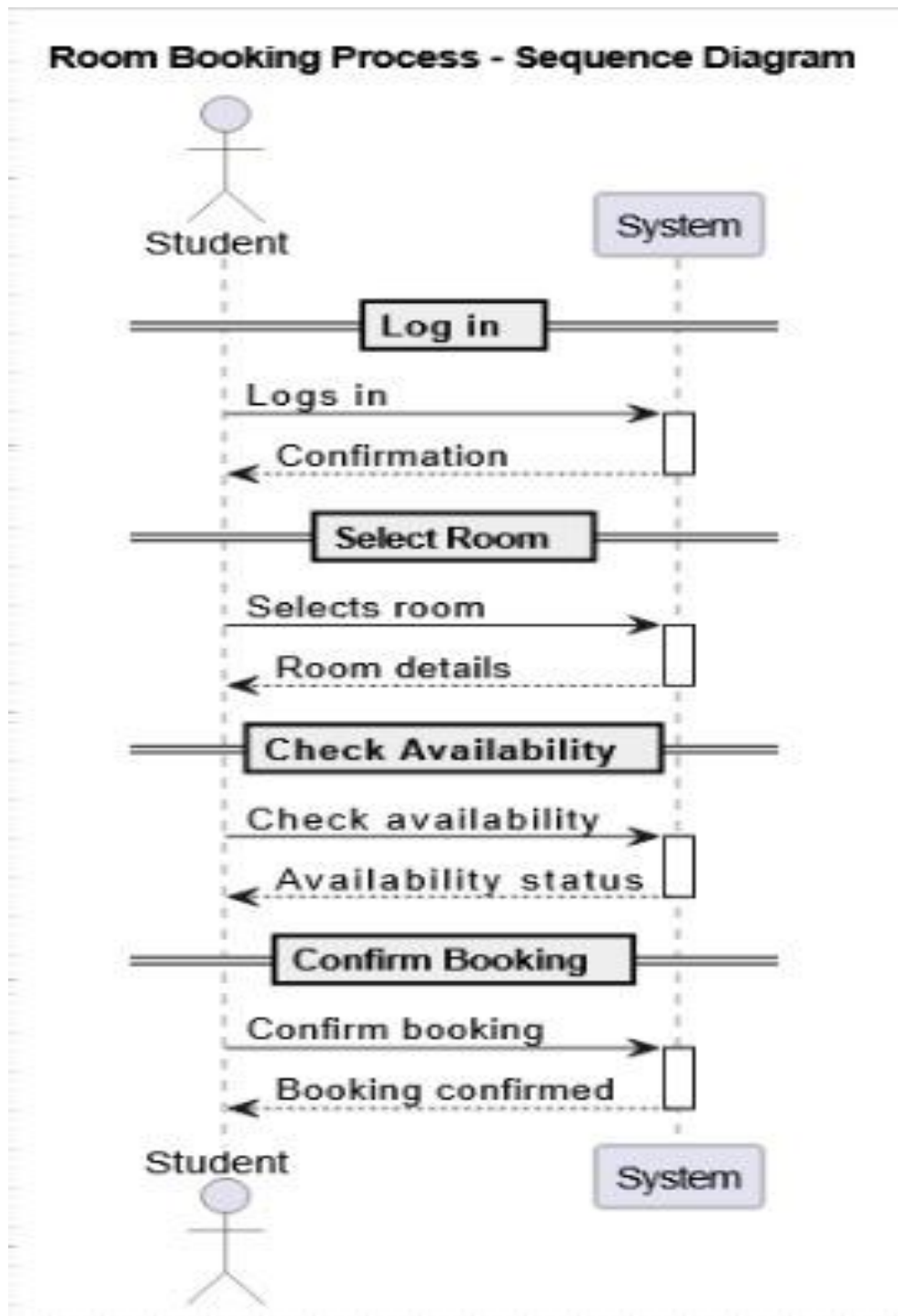


Figure 26 Room Booking Process - Sequence Diagram

Chapter Five

5. Implementation

Implementation is the do phase; an implementation is when all of the ideas and plans come together to create a working system. In this chapter, I will discuss the development of University Housing Management System and the tools that were used, or how we turned previously defined requirements into something that actually works.

This system provides electronic services which are energy-saving and easy to use for both students and administrators, such as room booking, maintenance management, complaint tracking etc. This web implementation process also leverages contemporary web-development technologies, with a primary focus on scalability and maintainability to ensure that the system remains relevant amid increasing user demands.

5.1 High and Low fidelity prototypes

Low fidelity prototype is low-tech concept where we use only pen and paper to sketch our ideas about design of interfaces. It's very easy concept which allow us to simply add or remove objects and finally to have guide for drawing high fidelity prototypes and on the end to convert it in real frontend interfaces.

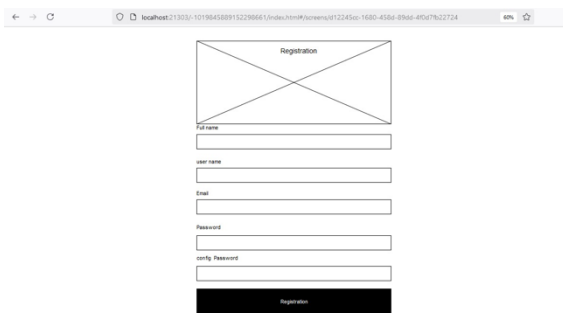


Figure 27 Low fidelity rehistration

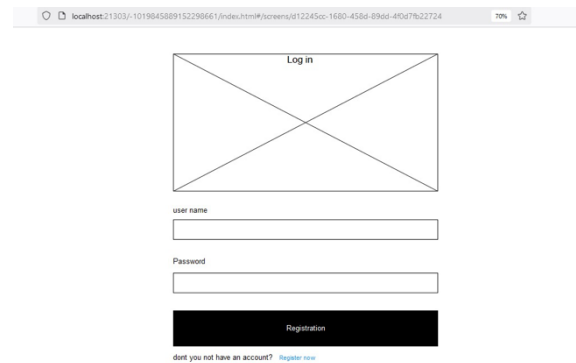


Figure 28 Low fidelity login

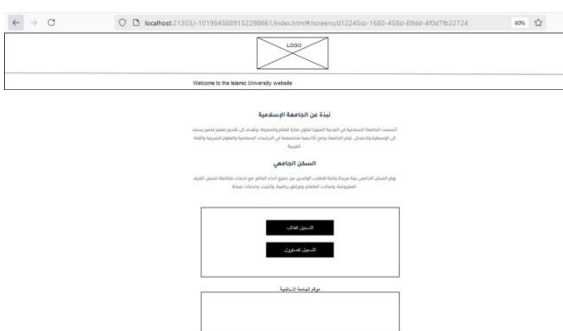


Figure 30 Low fidelity home

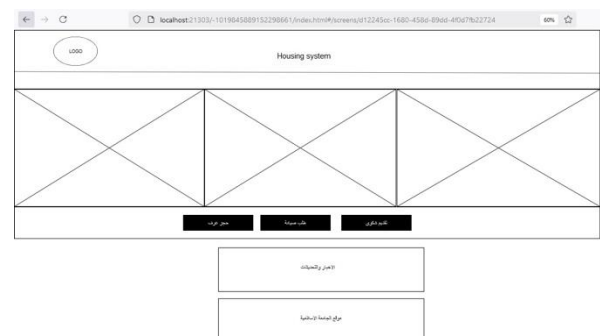


Figure 29 Low fidelity Service Page



Figure 31 High fidelity login



Figure 32 High fidelity registration

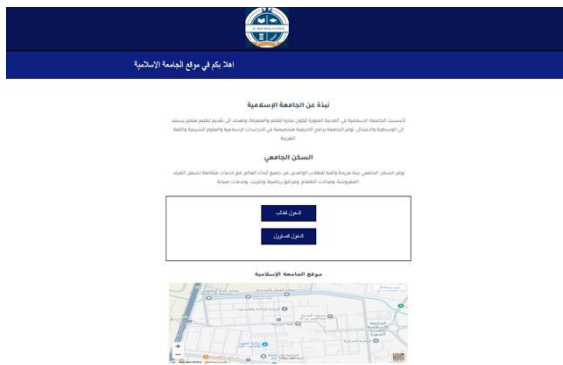


Figure 34 High fidelity home

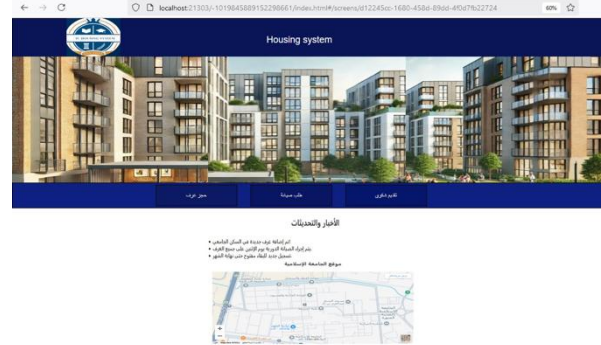


Figure 33 High fidelity Service Page

5.2 Tools and Technologies use in the Project

- **XAMPP** : Used as a local server to hook up the website to the database so you can handle pages and test them locally before deployment.
- **phpMyAdmin**: A database management tool used to control a website's database and the ability to call up data with a page (php), such as adding or modifying tables and queries.

5.2.1 Front-End

HTML: Used to build the structure of the web pages and design the user interfaces.

CSS: Used for any styling, sizing, formatting, and layouts of UI elements, text alignments, color changes, font changes.

JavaScript: Used to add interactivity to the website, dynamic effects or user interaction.



Figure 35 Front-End

5.2.2 Back-End

PHP: Used to link the front-end to the database, as well as process user requests, such as adding or modifying data in the database.

MySQL (via phpMyAdmin): Used to manage the database supporting the website, where user information, services, and other data are stored.



Figure 36 Back-End

5.3 Admin dashboard Page

The Admin's role on this dashboard is to fully manage and control the website. Based on what was presented in the picture, the admin's primary functions can be summarized as follows:

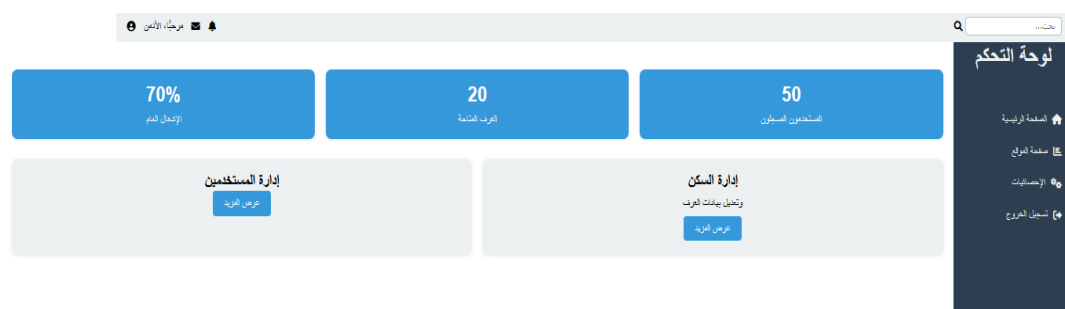


Figure 37 Admin dashboard Page

User Management:

Being able to add, edit, or delete users in the system.

Ability to observe users' data, such as name, role, and account status.

Room Management:

List the available rooms and update their information.

Provide the status of the rooms, such as occupied or available, and maintain the records accordingly.

Main dashboard: User management section & Room management section.

& Logging out of the dashboard.

Objective: This dashboard will present the Admin with the ability to monitor and manage all components of the website efficiently and easily for the effective operation of the system.

5.4 Home Page

The home page is the initial display of system welcome. It is designed in simple and aesthetically pleasing style reflecting the ease of use.

Main components:

Navigation bar: It helps the user to quickly move between sections of the system.

Intro - Short description about services provided by system.

Direct Links: Scoring to the most essential services like bookings, maintenance requests and complaints.

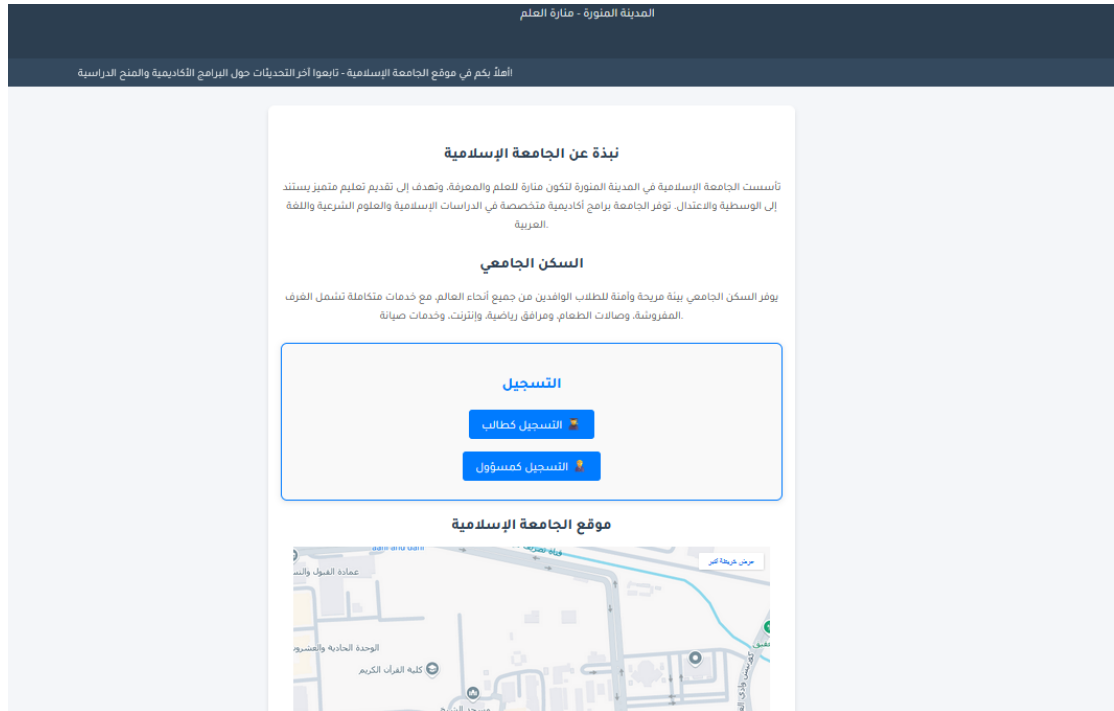


Figure 38 Home Page

5.5 Login Page

The login page is the entry point of security to the system.

Components:

Textboxes for inputting email/username and password

"Login" button.

In case of forgotten password, the link to retrieve it.

Features:

User interface that is intuitive.

Very secure use of encryption to protect login credentials.

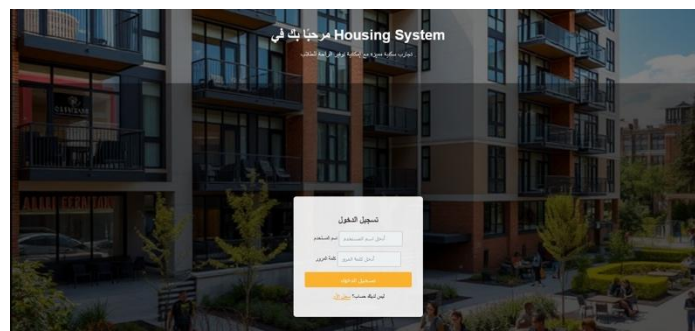


Figure 39 Login Page

5.6 Register

Allows you to register new users accounts to take advantage of the system services.

Components:

There will be text boxes where you can input your full name, student number, and email.

Choose a strong password.

Submit any documents that prove your income or place of residence, if needed.

Objectives:

Help introduce newly-positioned students to the system.

Perform data input validation and reliability.

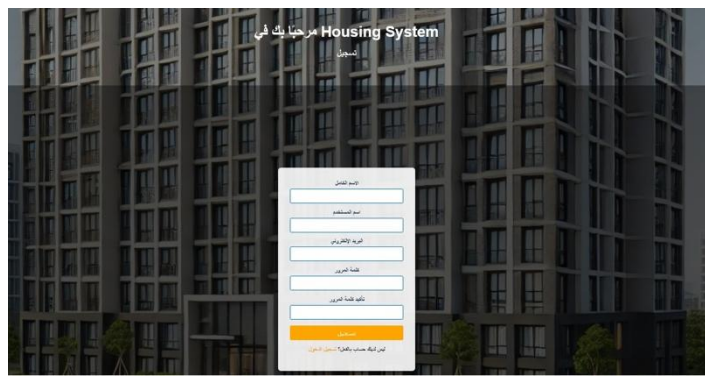
A screenshot of a web application titled "Housing System" with the subtitle "مرحبا بك في التسجيل" (Welcome to registration). The form is overlaid on a background image of a modern apartment building. The form contains several input fields: "الاسم الكامل" (Full Name), "رقم التسجيل" (Registration Number), "البريد الإلكتروني" (Email), "كلمة المرور" (Password), and "تأكيد كلمة المرور" (Confirm Password). Below these fields is an orange "التسجيل" (Register) button. At the bottom of the form, there is a small line of text: "نحن نؤكد على حماية بياناتك وسنحافظ على أمانها" (We ensure the security of your data and will maintain its safety).

Figure 40 Register

5.7 Service Page

The services page is the heart of the system where it helps the user to get access to all available services.

Available services:

Booking a room: Check for available and net the room.

Maintenance requests – Ascertain maintenance requests from rooms or facilities

Complaints: File and track complaints

Invoices and Payments: Options to view the invoice, as well as perform an electronic payment.

Design:

Interface strictly separated for each service, icons to make easy to identify options

Status Update Alert Messages (e.g. Reservation Confirmed & Maintenance Request Processed).



Figure 41 Service Page

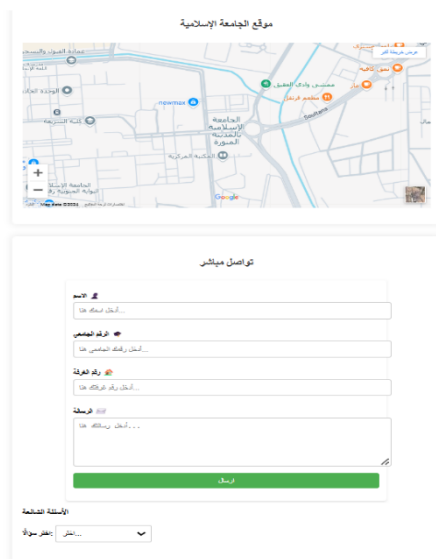


Figure 43 Service Page

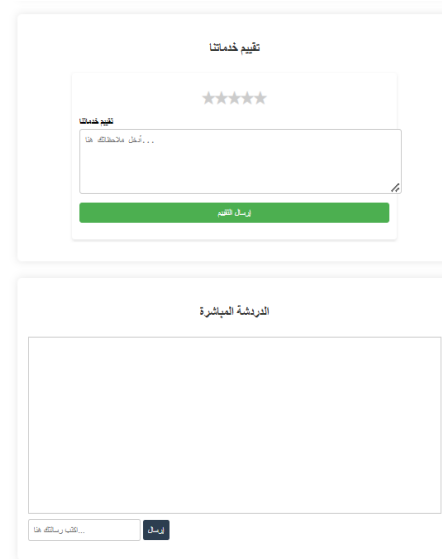


Figure 42 Service Page

5.8 Room Boking

It is an important part of the system where students can easily find and book free rooms.

Features:

Search options: Date, type of room and location / credit: AirBnBlock

Display available rooms information (capacity/price/facilities).

Finish booking: With a popup alert providing details of the accommodation, confirm the booking

Objectives:

Ensure seamless and effortless booking process
Make Room Management Process Efficient.



Figure 44 Room Boking

Chapter Six

6. Updates and improvements

After completing the first part of the project, we are now in the development and improvement phase of the website. In this chapter, we will present all the updates and modifications we made to the structure of the previous project. We developed the services in addition to improving the design and user interaction with all effectiveness. In the services section, we provided all possible services that the student can benefit from. In terms of design, we made modifications to attract the user and viewer of the site with ease and comfort.

We have updated the registration page (Figure 45) with a clear and easy format and provides input validation, ensuring that new users register quickly and accurately.

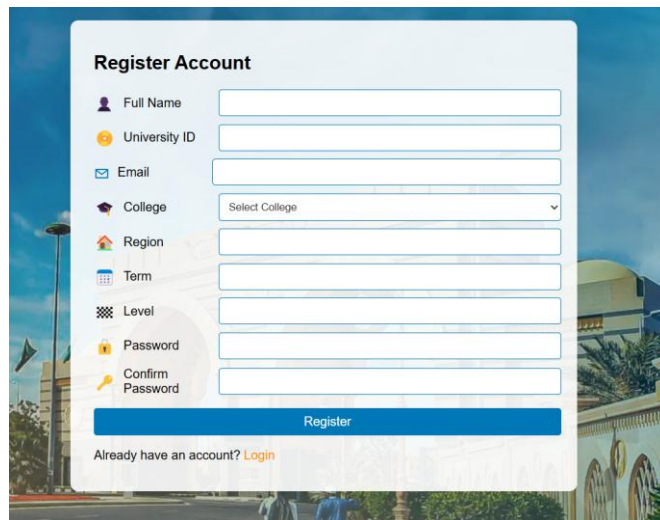
A screenshot of a web registration form titled "Register Account". The form is overlaid on a background image of a university building. It contains several input fields: "Full Name" (with a person icon), "University ID" (with a yellow ID card icon), "Email" (with an envelope icon), "College" (a dropdown menu with "Select College" text), "Region" (with a house icon), "Term" (with a calendar icon), "Level" (with a graduation cap icon), "Password" (with a lock icon), and "Confirm Password" (with a lock icon). A blue "Register" button is at the bottom. Below the button, it says "Already have an account? [Login](#)".

Figure 45 Register

The login page (Figure 46) now features a more user-friendly design with an alert for failed login attempts.

A screenshot of a web login page titled "Welcome to Housing System" with the subtitle "Experience outstanding accommodation with ultimate comfort for students." The background is a university building. A white login form is centered, containing "Student ID" and "Password" input fields (both with lock icons), a blue "Login" button, and a link "Don't have an account? [Register now](#)".

Figure 46 Login Page

On the services page (Figure 47), we have provided icons for each service, which makes it easier for the student to choose his service and carry out the required procedures afterwards.

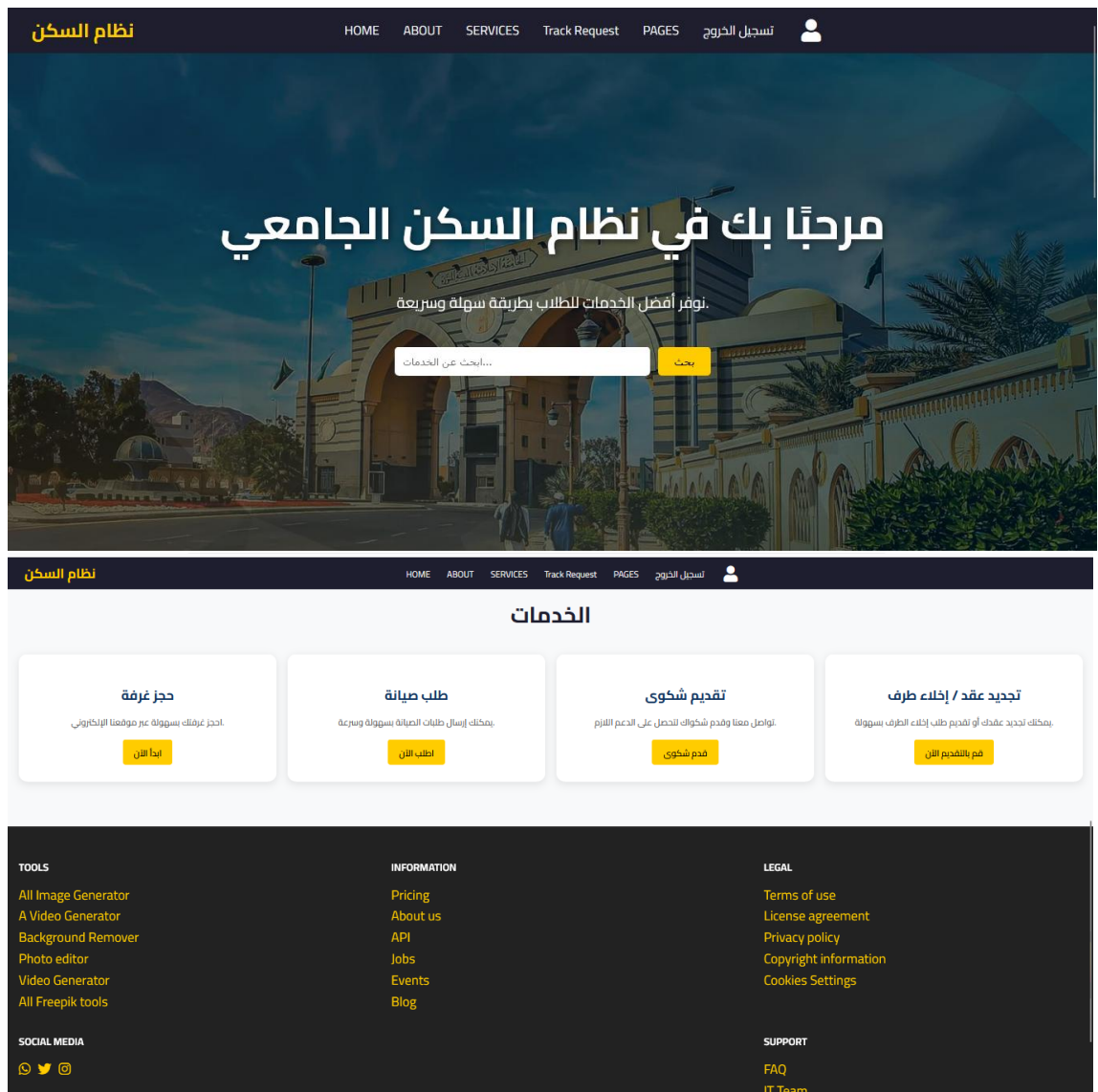


Figure 47 Service Page

The booking page (Figure 48) now includes the ability to check room availability in real time to avoid double bookings and improve allocation accuracy.

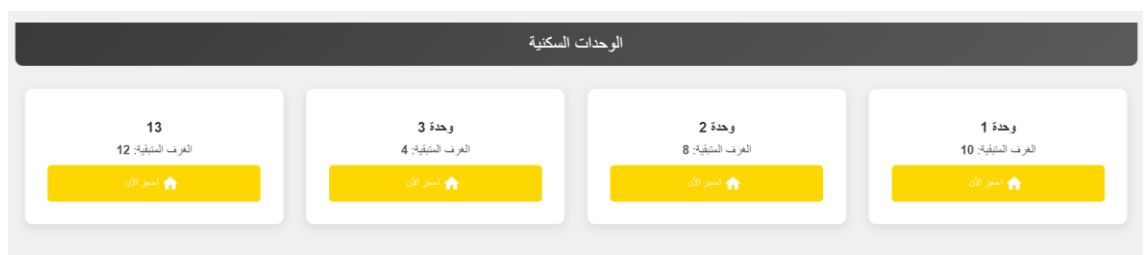


Figure 48 Booking Page

In the new payment page (Figure 49), we added the ability for the student to pay in several different ways that we provided for him.

Figure 49 Payment Page

The Contract Page (Figure 50) has been added to allow students to print their digital contracts and agree to the terms presented, reducing the need for paper forms.

Figure 50 Contract Page

The maintenance service has been improved and requests that students can request have been placed (Figure 51). Maintenance requests can also be tracked and the status can be tracked as well.

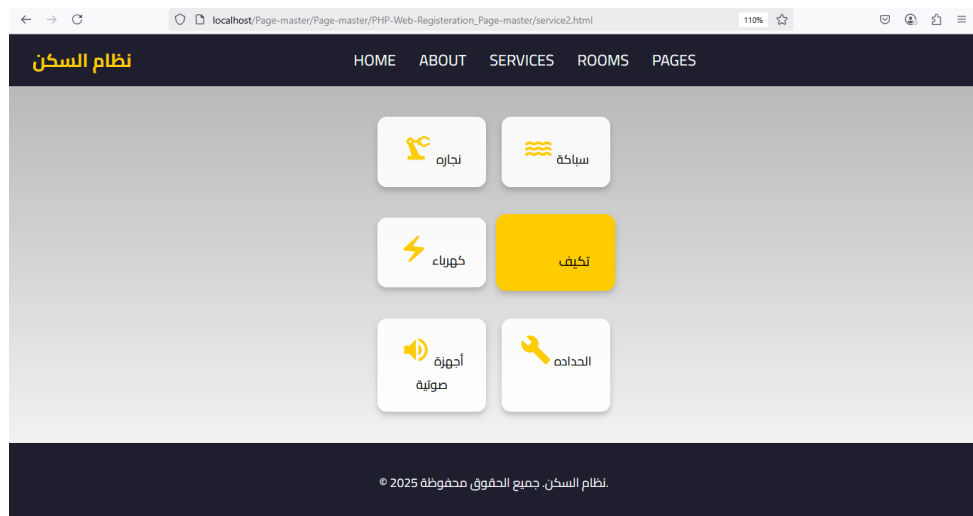


Figure 51 Maintenance Service

In Figure 52, the complaint form has been improved as shown. The student can choose the type of complaint, write the details, and upload a photo if possible.

The screenshot displays the 'نموذج الشكوى' (Complaint Form) interface. It features three selection buttons: 'مشرف' (Supervisor), 'طالب' (Student), and 'سكن' (Residence). Below these, there is a section for 'تفاصيل الشكوى:' (Complaint Details) with a text area for 'نوع الشكوى: مشرف' (Type of complaint: Supervisor). Another section for 'إرفاق صورة أو ملف:' (Attach photo or file) includes a file upload button labeled 'Browse' and a note '2 - Privacy.pdf'. At the bottom, there is a yellow button labeled 'إرسال الشكوى' (Send complaint).

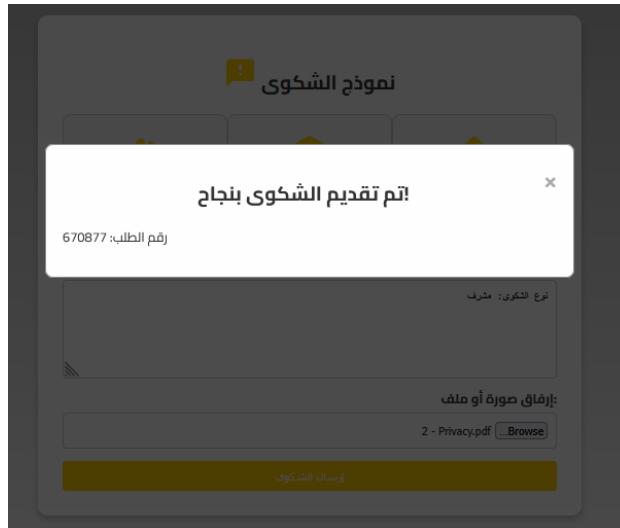


Figure 52 Complaint Service

Figure 53 gives the student the option to renew or vacate with ease, so that it also provides the university with ease in detecting students who have vacated housing.

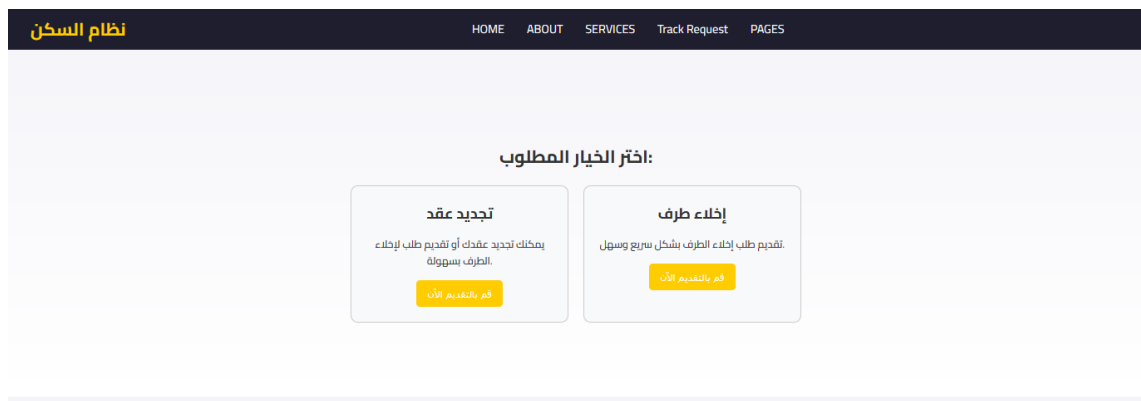


Figure 53 Renewal and evacuation

Figure 54 indicates that if the student chooses to renew the contract, a list appears for him to choose the term he wants and the required value is shown to him.

اختر الخيار المطلوب

تجديد عقد

يمكنك تجديد عقدك أو تقديم طلب لإلغاء الحظر بسهولة

قم بالتقديم الآن

إلغاء طرف

تقديم طلب إلغاء الطرف بشكل سريع وسهل

قم بالتقديم الآن

اختر الفصل الدراسي

ترم أول

قيمة الترم: 1000 ريال

الشروط والأحكام

عرض الشروط والأحكام ☐ أو أوافق على الشروط والأحكام

متابعة

الشروط والأحكام

- يجب دفع الرسوم في الوقت المحدد
- لا يمكن استرداد الرسوم بعد التسجيل
- يجب الالتزام بالقوانين واللوائح

اختر الفصل الدراسي

ترم أول

قيمة الترم: 1000 ريال

الشروط والأحكام

عرض الشروط والأحكام ☒ أو أوافق على الشروط والأحكام

متابعة

Figure 54 Renewal

In Figure 55, there is an evacuation list. The student fills out the form and attaches the papers. The application is officially submitted and the application number is shown to him.

The screenshot displays the 'نظام السكن' (Housing System) website interface. At the top, there is a navigation bar with links: HOME, ABOUT, SERVICES, Track Request, and PAGES. The main heading is 'اختر الخيار المطلوب:' (Select the required option:). Below this, there are two service cards: 'تجديد عقد' (Renew Contract) and 'إخلاء طرف' (Evacuation). The 'إخلاء طرف' card is highlighted in yellow and contains the text: 'تقديم طلب إخلاء الطرف بشكل سريع وسهل' (Submitting an evacuation request quickly and easily) and 'قم بالتقديم الآن' (Submit now). Below the cards, there is a section titled 'إخلاء الطرف:' (Evacuation:) with the instruction 'يرجى ملء النموذج في الأسفل لتقديم طلب إخلاء الطرف' (Please fill out the form below to submit an evacuation request). A text input field is present with the placeholder 'ط...' and a label 'عرض شروط الإخلاء' (View evacuation conditions). Below the input field, there is a 'Browse...' button with the text 'No file selected.' and a 'تقديم الطلب' (Submit request) button. A modal dialog box is overlaid on the page, displaying a success message: 'تم تقديم الطلب بنجاح! رقم الطلب: 976157 سبب الإخلاء: ظروف قاهرة' (Request submitted successfully! Request number: 976157 Reason for evacuation: Force majeure) with an 'OK' button. The background of the page is dark gray.

Figure 55 evacuation

Figure 56 update This page explains contract renewal and housing evacuation services. The student selects the service, and the form appears, along with the term selection. Each term has its duration and price. For example, the first term costs 1,000 riyals, costs 1,000 riyals, and lasts for months. The duration is determined with great precision based on the last day of the contract. The summer term also differs in that it lasts two months and is determined using the same mechanism.

طلب تجديد أو إخلاء

إخلاء السكن

تجديد العقد

سبب الطلب:

اختيار الترم:

الترم الأول - 1000 ريال

بتاريخ المقت:

2026-11-01

نهاية المقت:

2027-05-01

المدة:

6 أشهر

السعر:

1000 ريال

تأكيد التجديد

Figure 56 new evacuation and renewan

Figure 57 On this page, the service of monitoring requests related to maintenance and complaints has been explained. Here, each user can track his requests only with the number and clarify the date, as well as view the recorded images. A communication method has been added so that the status of the request is determined by the admin or supervisor and it is responded to in the table or appropriate place so that the student will know when he will be contacted.

نظام السكن
HOME ABOUT SERVICES Track Request PAGES

مراقبة طلبات الصيانة والشكاوى

طلبات الصيانة

الرسالة	الحالة	التاريخ	الصورة المرفقة	تفاصيل الطلب	التحويلة	موقع البلاغ	رقم الطلب
لا توجد طلبات صيانة حاليا							

الشكاوى

الرسالة	الحالة	التاريخ	الصورة المرفقة	تفاصيل الشكاوى	نوع الشكاوى	رقم الشكاوى
لا توجد شكاوى حاليا						

نظام السكن - جميع الحقوق محفوظة 2025 ©

Figure 57 monitoring requests

Figure 58 and 59 This page has been updated and added to determine the priority according to the population of the Kingdom of Saudi Arabia region, so that priority is given according to the applicable proximity to the city. The mechanism for this process is to compare the registered city with Medina, and then it is determined and the person is allowed to complete the payment process if he is far away, for example (Al-Khobar or Dammam).

صفحة الدفع

تفاصيل الحجز:

username : 421709663
UserID : 47
region: dammam

طريقة الدفع:

☐ بطاقة التمان

☐ مستفيد من الضمان الاجتماعي

إتمام الدفع

أنت مؤهل لإتمام الدفع

2025 | جميع الحقوق محفوظة © Housing System

Figure 58 New Payment page(1)

صفحة الدفع

تفاصيل الحجز:

username : 421709663
UserID : 47
region: medina

طريقة الدفع:

☐ بطاقة التمان

☐ مستفيد من الضمان الاجتماعي

إتمام الدفع

ليس مؤهلاً للدفع لأن المدينة قريبة

2025 | جميع الحقوق محفوظة © Housing System

Figure 59 New Payment page(2)

In Figure 60 here, we show and explain to you our database, the mechanism for its installation, what it contains, and how the tables are linked to each other. We also explained several users to clearly explain the mechanism of the system's work.

The screenshot displays the phpMyAdmin web interface. The top section shows the login form with fields for username (root), password, and server choice (MySQL). Below the login form, there is a table listing the databases and their characteristics.

الجدول	الحجم	التخزين	النوع	العملية
bookings	30 بايت	utf8mb4_0900_ai_ci	MyISAM 2	إضافة
items	16.0 كيلوبايت	utf8mb4_general_ci	InnoDB 0	إضافة
room_bookings	32.0 كيلوبايت	utf8mb4_general_ci	InnoDB 10	إضافة
units	2.8 كيلوبايت	utf8mb4_0900_ai_ci	MyISAM 32	إضافة
user	16.0 كيلوبايت	utf8mb4_general_ci	InnoDB 17	إضافة
المجموع	70.9 كيلوبايت	utf8mb4_0900_ai_ci	MyISAM 61	

Below the table, there is a section for creating a new table, with fields for the table name and the number of columns.

The bottom section shows the table structure for the 'bookings' table, with columns: level, term, region, college, password, email, username, name, id. The table is sorted by 'id' in ascending order.

Figure 60 Database

Figure 61 Admin services are easily activated, as shown on the pages.

- Room management includes room requests. Students who have successfully reserved rooms are shown, along with the room number and username to identify the student. The admin confirms the student's status to allow them to use the available services. The admin may reject the student's status if problems arise, and services are blocked or hidden from the student. The student's status is shown. The admin and the student are linked through a table called "Status." This allows the student to know whether they have been accepted into the system or not.

Admin Dashboard		Room Management			
Student requests		Room Number	User Name	Term	Status
Units		9		580282546	Accept
Settings		1		56866990	Accept
		3	bader ahmed	0	Accept
		1	rs	1	Accept
		1	r mmdoh	8	Empty
		1		568660808	Empty
		8	jaber saih	0	Empty
		6		788889999	Empty
		12	zoro	590282564	Accept
		3	omir	111111111	Accept

Figure 61 Admin Room Management

Figure 62 It also includes monitoring student requests in the complaints and maintenance department. They are also contacted by determining the status of the request, which is controlled by the admin by choosing 4 statuses. The first (Ambitious) is the initial status of the request. The second (Accepted) is a message that the student will be contacted in a short time. The third status (Not Accepted) is the request being rejected by the admin and the student is shown to try again later. After completing the process, the fourth status (Process Complete) is determined and a message appears to the student that he has been contacted, the process resolved and completed. This facilitates and provides a faster process for communication between them.

الشكاوى							
رقم الشكاوى	نوع الشكاوى	تفاصيل الشكاوى	الصورة المرفقة	اسم المستخدم	التاريخ	الحالة	تغيير الحالة
440097	سكن	I	عرض الملف	421009883	2025-04-27 20:19:33	empty	empty تغيير
727249	طالب	II	عرض الملف	421009883	2025-04-27 20:20:04	empty	empty تغيير
479131	طالب	II	عرض الملف	421009883	2025-04-27 20:20:38	empty	empty تغيير
576832	سكن	I	عرض الملف	421009883	2025-04-27 20:20:51	empty	not accept تغيير
269167	سكن	I	عرض الملف	421009883	2025-04-27 20:22:54	empty	empty تغيير
298004	سكن	تفاصيل الشكاوى عن: سكن	عرض الملف	421009883	2025-04-27 20:23:03	empty	accept تغيير
664198	مشرف	تفاصيل الشكاوى عن: مشرف	عرض الملف	421014553	2025-04-27 20:27:40	empty	The operation is complete تغيير
822848	طالب	تفاصيل الشكاوى عن: hh طالب	عرض الملف	999999999	2025-04-27 20:29:53	accept	accept تغيير
708388	طالب	تفاصيل الشكاوى عن: طالب	عرض الملف	421009883	2025-04-27 22:55:43	not accept	not accept تغيير

Figure 62 Admin monitoring student requests

Figure 63 This page explains the management of the residential units in the system and is controlled by the admin. Its name, the number of rooms available, and the number of rooms reserved are shown. The admin can modify, delete, and add if he wants or if the administration asks him to, while ensuring that it is displayed correctly. This provides ease when a problem occurs in one of the units.

إدارة الوحدات السكنية

الاسم	عدد الغرف	عدد الغرف المحجوزة	الإجراءات
unit 1	50	26	تعديل حذف
unit 2	20	19	تعديل حذف
unit 3	30	2	تعديل حذف

إضافة وحدة سكنية جديدة

اسم الوحدة:

عدد الغرف:

عدد الغرف المحجوزة:

[إضافة وحدة سكنية](#)

Figure 63 Admin Management Units

1. `password_verify()` & `password_hash()` : This function verifies the validity of the entered password and its encryption method: `if (password_verify($password, $user['password']))` Here, it checks whether the entered password matches the encrypted password stored in the database, as shown Figure 64.

- Example 1: `password = password_hash($password_1, PASSWORD_DEFAULT).`
- Example 2: `f (password_verify($password, $user['password'])).`

- password

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...hashedpassword1\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$J1EPeVig7OFpc35XuGVVvekFTQl9ZGkZFcnefbZ590F

...\$2y\$10\$ScdKOUNxtykvNdX54rXMbedJxJ0ajr9WgX.bjftRYCZ

...\$2y\$10\$DqLF18FVOGucYz/bNH2wmuP.Ba.6yk63gFNfGLt4Wkt

...\$2y\$10\$z2qHVgjZY.HFcndDkJkJbh.w/W/CZI4Izbtts.qWrdJAC

...\$2y\$10\$9LNRfKfZlY137mQMqQ5wXv.q427qPY2D6fAB/OdE5XOm

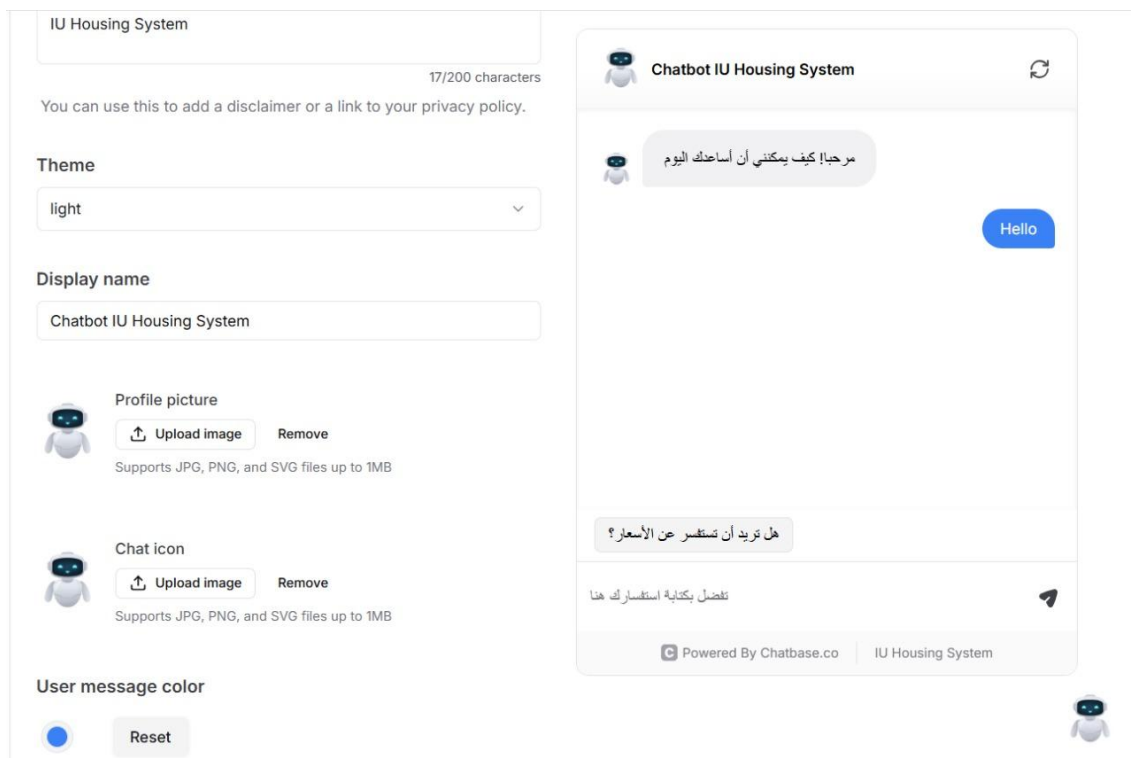
{ 59 }

6.2 Chat bot

With the advancement of technology and websites, the chatbot feature has become essential on every website, and indeed one of the most important features of websites today. Based on our project, we designed a fully integrated chatbot that adapts to the website and its questions only.

Steps to create the bot:

We used the Chatbase platform to develop smart bots. The process begins by creating a smart agent within the platform, specifically for the student housing website. This agent is programmed solely for the website's questions, providing the agent with the full text of the question and the answer. Integrating the bot with the website is done through JavaScript code provided by the platform.



Create new agent

Files

Text

Website

Q&A

Notion

Text

الإجابة: نعم، يوجد دعم فني متاح عبر قسم الدعم داخل النظام. يمكنك تقديم طلب دعم للحصول على المساعدة في حل واجهيت أي مشكلة.

22. هل يمكنني الوصول إلى النظام من خلال الهاتف المحمول؟
الإجابة: نعم، يمكنك الوصول إلى النظام من خلال المتصفح على هاتفك المحمول أو عبر تطبيقات الهواتف الذكية إذا كانت متاحة.

23. هل هناك رسوم حجز؟ نعم هناك رسوم في بداية الحجز وتكون ب 500 وهي مستردة بالطبع في.

إجابة لأي سؤال خارج المواضيع المحددة:
الإجابة: "المعلومات غير متاحة حاليًا. إذا كان لديك أي استفسار آخر حول نظام السكن الجامعي أو كيفية استخدامه، لا تتردد في سؤالي"

1. How can I register in the system?

Answer: You can register through the registration page by entering your personal information such as name, email, and student number. After that, you will be asked to set a password to create your account.

2. What information do I need to book a room in the university housing?

Answer: You need to log in first. Then, you can select an available room based on your desired booking dates and duration. Additional details such as whether you're a

12 KB • 8,910 characters

Connect

Embed

Share

Integrations

Embed

Embed a chat bubble **Recommended**

☒ Embed a chat bubble on your website. Allows you to use all the advanced features of the agent. Explore the [docs](#).

☐ Embed the iframe directly

Add the agent anywhere on your website

Configuration

On the site

www.chatbase.co

```
1 <script>
2 (function(){if(!window.chatbase||window.chatbase("getState")!= "initialized"){window.chatbase=([...arguments])=>{if(!window.chatbase.q){window.chi
3 </script>
```

Figure 65 chat bot operations

6.2 Challenges and Issues

During the development phase, we encountered several challenges:

- Difficulty integrating instant updates between the front-end and database due to server limitations.
- Designing a unified interface that works across different devices and mobile phones and is compatible with all screen sizes.

6.3 Conclusion

The conclusions: The project University Housing System is a significant turn around in categorizing housing management in What Happened at Islamic University. By migrating the old paper based system to the advanced electronic system we were able to solve a whole bunch of problems students and administrative staff had in the past like challenges writing requests for room booking, maintenance reporting, complaint tracking, resolution and report writing challenges. Tasks that would have previously required significant time and effort can now be accomplished quickly and efficiently, and resources can be better utilized by all parties.

But just as with every major project, we have faced a few hurdles, notably how to implement real-time updates between the front-end and the database, and ensure that the entire system is compatible with all devices and satisfactory for all user needs. We overcame many of these challenges the best we could, and everything we did encounter we learned from along the way. We still have to work on improving these things.

With all these challenges, we think that we were able to give the university a powerful and well-needed solution that simplified the system for the students and for the administrators. Not to mention the fact that it is not just a solution to the current issues - it is a platform for the ongoing development and innovation around the university services. We are excited by the possibilities created by the potential opportunity to further develop this system, including this becoming integrated into additional university systems and grow the functionality it provides to support our increasing number of students.

In other words, this project serves as a major milestone in the ongoing digital evolution of university services that will open new horizons to IT based solutions. We believe that this system will be a blueprint to enhance services of the university using current technologies and its future implementations.

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